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Non-intrusive tracking of objects and fluids in wellbores

Abstract

Systems and methods of the present disclosure relate to non-intrusively monitoring a fluid level or an object in a conduit. A system comprises a component positioned to control flow into or out of the conduit to induce pressure waves in the conduit; a pressure transducer in fluid communication with the conduit, the pressure transducer positioned to measure pressure responses in the conduit due to contact of the pressure waves with the fluid level or the object; and a system controller operable to: receive pressure data from the pressure transducer, the pressure data comprising the pressure responses; and determine a distance of the fluid level or the object in the conduit, relative to the component or the pressure transducer, based on the pressure responses.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage application, filed under 35 U.S.C. 371, of international patent application no.

BACKGROUND

(1) In the oilfield, knowledge of a well condition such as a fluid level, for example, may facilitate optimal production operations. Existing technologies may employ various sensors such as pressure gauges or density gauges for determining the fluid level. However, implementation of these gauges may be limited by distance. Therefore, these gauges may only be installed at particular depths in a well. Additionally, older wells may not include these gauges which may limit techniques to determine the fluid level therein.

(2) Although well intervention may be another option to determine the fluid level in the well, this type of operation may not allow for a continuous monitoring of the fluid level throughout a life cycle of the well. Further, the well intervention may involve a substantial amount of time, resources, or risk. For example, servicing an offshore installation such as a normally unmanned installation (NUT), may require a mobilization of a drilling rig.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These drawings illustrate certain aspects of some examples of the present disclosure and should not be used to limit or define the disclosure.

(2) FIG. 1A illustrates a wellbore including a target to be located, in accordance with examples of the present disclosure;

(3) FIG. 1B illustrates an annulus of a wellbore including a target to be located, in accordance with examples of the present disclosure;

(4) FIG. 1C illustrates an underground fluid storage facility, in accordance with examples of the present disclosure;

(5) FIG. 2 illustrates a flow chart for locating and tracking of targets such as moving or stationary objects and/or fluids within conduits, in accordance with examples of the present disclosure;

(6) FIG. 3 illustrates a pressure response during a tracking of targets disposed in a conduit, in accordance with examples of the present disclosure; and

(7) FIG. 4 illustrates a measured pressure response with noise, in accordance with examples of the present disclosure.

DETAILED DESCRIPTION

(8) The present disclosure generally relates to real-time non-intrusive techniques for monitoring targets such as fluid levels or objects within conduits such as a wellbore or an underground storage facility. The targets to be tracked may include various objects or fluids such as hydrates, media or phase change of fluid(s), fluid influxes into a wellbore, and/or fluid levels in gas storage wells or cavern storage facilities that may store various fluids such as hydrogen. In some examples, the hydrates may be tracked during hydrate remediation operations. The targets may also include downhole plugs, tools, or debris disposed in the wellbore or an annulus thereof.

(9) In particular examples, a system controller and a pressure measurement device may be implemented into existing infrastructure that may include components such as, for example, a wellhead, piping, a valve, a pump, or a vessel. The existing infrastructure may be utilized to control a flow of a fluid to induce positive and/or negative pressure waves within the conduit. Each of the components may be positioned to control a flow of fluid within a conduit to induce pressure waves.

(10) Tracking of a target may occur in real-time via an analysis of induced pressure waves and their corresponding responses that travels along the conduit back to the origin or source of the pressure wave inducement. The tracking may be performed by inducing pressure waves in the conduit, and analyzing pressure responses to determine real-time locations, such as a depth or a distance of the

target within the wellbore, relative to a position of the source of the pressure wave inducement. Location updates or location information such as fluid levels or object position may be transmitted to a web portal/or remote hardware to provide live tracking.

(11) In some examples, at least two pressure waves may be induced within a wellbore to elicit or cause at least two corresponding pressure responses that may reflect off of the desired target that is disposed within the wellbore, and travel as a pressure response back along the wellbore to the source of the pressure wave inducement for analysis by a system controller. An interaction due to contact between the induced pressure waves and the desired target may result in the pressure responses. The system controller may determine a location of the target based on the pressure wave responses.

(12) An automated system, without human involvement, may adjust a production parameter such as a flow rate into or out from the wellbore; reduce a water load in the wellbore (e.g., add a foaming agent into the wellbore); or adjust a valve such as a choke, all of which may be based on the pressure responses. In some examples, the real time tracking of fluid levels in a production well may provide insight for reservoir analysis.

(13) In certain examples, the real-time tracking may allow for an accurate assessment of time duration for cementing operations (e.g., tracking positions of plugs) and fishing operations for downhole tools that may be stuck in the wellbore. Additionally, the real-time tracking may allow for locating of objects or fluids in an annulus that may include an open-hole portion.

(14) The techniques described herein may provide for increased accuracy and distance capability over other methods such as acoustic diagnostics, fiber optic downhole gauging or fiber optic cables. For example, numerous sensors positioned along a conduit, are not required; rather, flow rates and pressures in the conduit may be measured at or near the wellhead, using one or more pressure waves. Additionally, locations may be overlaid upon wellbore completion diagrams.

(15) FIG. 1A illustrates a target **100** disposed in a conduit such as a wellbore **102** that extends into a subterranean formation **104**, in accordance with examples of the present disclosure. In some examples, the target **100** may include various objects and/or fluids. For example, the target **100** may include debris, a top of a cement column, plugs, downhole tools (e.g., bottom hole assembly including measurement modules), completion collapse, various fluids, and/or fluid changes (e.g., based on density or viscosity) in the wellbore **102**. In some examples, the techniques as described herein, may non-intrusively provide a depth of a top of downhole tool, such as a stuck tool, allowing for an effective fishing operation.

(16) The wellbore **102** may include conduits **106** such as casing which may include a first section **108**, a second section **110**, and a third section **112** concentrically disposed within the wellbore **102**. Each section **108**, **110**, and **112** of the casing may include a corresponding valve **114** located at a wellhead **116** (or adjacent pipes in fluid communication with the wellhead **116**). The valves **114** may allow for an ingress or egress of fluid into or out of the wellbore **102** to induce a pressure wave **118**.

(17) Manipulation such as rapid opening and closing of a valve **114** may induce pressure waves within the wellbore **102**. The valve **114** may open and close within seconds. In some examples, 5 seconds may be the maximum open time. In other examples, the valve **114** may remain open for 1 second or less before it is closed. Longer durations may work as well (subject to the system parameters and depth of the object/fluid position).

(18) Upon manipulation of a valve **114**, the pressure wave **118** may travel inside and along the corresponding conduit **106** (e.g., at the speed of sound) and may reflect off the target **100** as a reflected pressure wave **120** to travel back to the valve **114**. The reflected pressure wave **120** may correspond to features detected in the wellbore **102**, such as the target **100**, for example. The reflected pressure wave may be considered a pressure response in some examples.

(19) The pressure wave **118** may include a positive and/or a negative pressure wave that may be induced within the wellbore **102**. A number of suitable techniques, in addition to the valve **114**,

may be used for inducing pressure waves. For example, a reciprocating pump **128** may remove fluid from the wellbore **102** to induce negative pressure waves. The pumping may occur in intervals. In additional examples, existing pressure waves in the system may be used as a source for the pressure recording and data collection. In particular examples, the valve **114** may be a hydrodynamic valve that may be operated manually. In other examples, the valve **114** may be automated and/or controlled remotely. By way of further examples, fluid may be released from the wellbore **102** into an expansion vessel **130**, or the valve **114** may be used to bleed fluid from the wellbore **102**.

(20) In addition to or in combination with the negative pressure, a positive pressure wave may also be induced in the wellbore **102**. In particular examples, a supply tank or a supply vessel **132** may supply fluid into the wellbore **102** to induce a positive pressure wave in the wellbore **102**. In some examples, the supply vessel **132** may utilize a pump **134** to move fluid into the wellbore **102**. The supply vessel **132** may provide any suitable fluids into the wellbore **102**. Additionally, the pump **128** may pump fluid into the wellbore **102**. Non-limiting examples of fluids passed into the wellbore **102** may include gas such as nitrogen, carbon dioxide, and/or natural gas, into a gas system, or water (or other suitable liquid) into a liquid system. In some examples, hydrodynamic waves may be induced by an injection of the gas into the wellbore **102**.

(21) The reciprocating pump **128** and the vessels **130** and **132** may be in fluid communication with the wellbore **102** via valves **114** and/or conduits **136**. It should be noted that in some examples, the mechanisms to induce pressure waves such as those described herein, may be disposed at an above-ground location such as at a surface of the wellbore **102**. The pressure waves may be induced at regular intervals, or the intervals may be variable. In particular examples, sonic waves may be induced at regular or variable intervals and may be utilized in accordance with examples of the present disclosure.

(22) The pressure transducer **122** may be in fluid communication with the wellbore **102** to measure pressure variations at a high or ultra-high sampling rate (e.g., 1 kilohertz (kHz), 1-4 kHz, or greater than 4 kHz). In certain examples, the sampling rate may be less than 1 kHz or greater than 4 kHz. In some examples, the pressure transducer **122** may measure pressures up to (but not limited to) about 22,500 pounds per square inch (psi) or about 1550 bar. Signals from the pressure transducer **122** may be recorded with a system controller **124**.

(23) Once the data is collected, analyzed, and/or extrapolated into profiles, for example, via the system controller **124**, a diagnostic of the well condition may be provided. In some examples, this may include reporting to facilitate decision-making and remediation plans without sacrificing production uptime or throughput. In some examples, the system controller **124** may be operated remotely (e.g., wirelessly or connected by a cable) via a device **126** which may include a personal computer, tablet, smartphone, or other digital device.

(24) In some examples, pressure wave inducement in the wellbore **102** may occur manually or may be automated via the system controller **124**. The system controller **124** may control fluid flow into and out of the wellbore **102** based on the pressure responses or the fluid levels in the wellbore **102**. For example, the system controller **124** may be operable to control various components such as the valves **114** and/or the pumps **128** and **134**, to induce the pressure waves that may travel through the wellbore **102** eliciting a pressure response. The pressure response includes a reflected pressure wave that has been reflected off the target **100** back to the source of the pressure wave inducement. The reflected pressure response may be measured by the pressure transducer **122** and recorded at a high or ultra-high rate (e.g., at least 1 kHz) by the system controller **124** for analysis, such as determining fluid levels within the wellbore **102**, pressure profiles of single phase and multi-phase fluids in the wellbore **102**, and/or determining a location of an object in the wellbore **102**.

(25) Upon manipulating a component (e.g., a valve, pump, and/or vessel) of a hydraulic system, a fluid flow may temporarily be stopped or restricted to induce a pressure wave in the wellbore **102**, and a pressure in the wellbore **102** may be continuously recorded at a point upstream to the

component, using the Joukowski equation, for example:

$\Delta p_{\text{sub.a}} = \rho u a$ (1) where $\Delta p_{\text{sub.a}}$ represents a surge pressure; ρ represents a fluid density, u represents a fluid flowing velocity and a represents the speed of sound in the fluid, to estimate the magnitude of the water hammer and using the Darcy-Weisbach equation:

(26) $p_f = \left(\frac{f}{2}\right)\left(\frac{L}{d}\right) u^2$ (2) where f is the friction factor, L is a pipe length, d is a pipe diameter, ρ is fluid density and u is fluid velocity, to determine the frictional pressure drop, thereby obtaining a time-log of the pressure change in the conduit. A distance-log of pressure change may be obtained from the time-log and an estimate of the speed of sound in the actual multiphase flow media, using the formula:

$\Delta L = 0.5a\Delta t$ (3) to obtain the relation between time (Δt) and distance (ΔL). This technique may allow for monitoring of fluids or objects within a conduit, for example. In some examples, a system controller may determine the location of a target in a wellbore relative to a location of a measured pressure response or a location of the pressure wave inducement via Equations 1 to 3. For example, the system controller may calculate a distance from a pressure transducer to the target as half the distance a pressure wave travels from the time of the pressure wave inducement to the time the pressure response is measured or received by the pressure transducer. The distance between the pressure transducer and the target may be utilized to calculate a distance to the target relative to a pressure inducement location such as a valve or pump, for example.

(27) The system controller **124** may include a display, a storage unit, and/or any instrumentality or aggregate of instrumentalities operable to compute, estimate, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, the system controller **124** may be a computer, a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The system controller **124** may include a processing unit (e.g., microprocessor, central processing unit, programmable logic controller (PLC), etc.) that may process data by executing software or instructions obtained from a local non-transitory computer readable media (e.g., optical disks, magnetic disks). The non-transitory computer readable media may store software or instructions of the methods described herein. Non-transitory computer readable media may include any instrumentality or aggregation of instrumentalities that may retain data and/or instructions for a period of time. The non-transitory computer readable media may include, for example, storage media such as a direct access storage device (e.g., a hard disk drive or floppy disk drive), a sequential access storage device (e.g., a tape disk drive), compact disk, CD-ROM, DVD, RAM, ROM, electrically erasable programmable read-only memory (EEPROM), and/or flash memory; as well as communications media such wires, optical fibers, microwaves, radio waves, and other electromagnetic and/or optical carriers; and/or any combination of the foregoing. The system controller **124** may also include input device(s) (e.g., keyboard, mouse, touchpad, etc.) and output device(s) (e.g., monitor, printer, etc.). The input device(s) and output device(s) provide a user interface. For example, the system controller **124** may enable an operator to select and perform analysis, view collected data, view analysis results, and/or perform other tasks.

(28) The system controller **124** may be in communication (e.g., wire or wireless) with various components via various communication paths and may be operable to control the components. In some examples, the system controller **124** may be operated remotely (e.g., wirelessly) via a device **126** which may include a personal computer, tablet, smartphone, or other digital device. In some examples, the system controller **124** may be battery-powered (e.g., rechargeable lithium-ion battery or other type of batteries) with up to 15 hours (or more) of operation time and may include piezoelectric switches. In other examples, the system controller **124** may be powered by an electric grid and/or a portable generator.

(29) FIG. 1B illustrates the target **100** disposed within an annulus **144** that extends into the

subterranean formation **104**, in accordance with examples of the present disclosure. The target **100** may be disposed in the annulus **144** between the first section **108** and the second section **110** of casing, in some examples. In some examples, the target **100** may include debris from an un-cased section or an open-hole interval **150** that may be in fluid communication with the annulus **148**.

(30) Upon manipulation of a valve **114**, a pressure wave **152** or wave may travel inside and along the sections **108** and **110** in the annulus **144** (e.g., at the speed of sound) and may reflect off the target **100** as a reflected pressure wave **154**. The reflected pressure wave **154** may correspond to features or fluids detected in the annulus **148**, such as the target **100**. In certain examples, the system controller **124** may manipulate the valves **114** based on fluid levels within the annulus **144**.

(31) The pressure transducer **122** may be in fluid communication with the annulus **144** to measure the pressure variations therein. Signals from the pressure transducer **122** may be recorded with the system controller **124**. As previously noted, the system controller **124** may be operable to control the valves **114**, the pump **128**, the pump **134** that is fluidly coupled to the supply vessel **132**, and/or the valves **114** to induce the pressure waves that may travel through the wellbore **102** eliciting a pressure response. This data may be collected, analyzed, and/or extrapolated into profiles via the system controller **124** to provide a diagnostic of the well condition.

(32) In some examples, the system controller **124** may be operated remotely (e.g., wirelessly) via a device **126** which may include a personal computer, tablet, smartphone, or other digital device suitable for surface and underwater. It should be noted that the configurations as shown on FIGS. **1A** and **1B** may be utilized onshore or offshore and may also be applicable to underground caverns or storage facilities.

(33) FIG. **1C** illustrates an underground storage facility **160**, in accordance with examples of the present disclosure. The underground storage facility **160** (e.g., a salt cavern) may be disposed in a subterranean formation **164**. Various types of fluids may be stored within the underground storage facility **160**. Non-limiting examples of the fluids may include a liquid and/or a gas such as brine, natural gas, and or hydrogen.

(34) A first conduit **166** and a second conduit **168** may extend into the cavern **162** from a location **170** that may be above the ground **172** (e.g., a surface location). The conduits **166** and **168** may pass liquids or gas into or out from the underground storage facility **160**.

(35) For example, a gas **175** may flow into the underground storage facility **160** via the first conduit **166**. A valve **114** may be operable to control the flow of the gas **175** into the underground storage facility **160**. The gas **175** may flow into the underground storage facility **160** from a fluid source **182** such as a pipeline or a vessel. In some examples, a liquid **183** may flow via a second valve **114** from the underground storage facility **160** into a structure **184** such as a pipeline or a vessel.

(36) As previously noted, manipulation such as the rapid opening and closing of a valve **114** may induce pressure waves within the underground storage facility **160**. The valve **114** may open and close within seconds. In some examples, 5 seconds may be the maximum open time. In other examples, the valve **114** may remain open for 1 second or less before it is closed. In other examples, longer durations may work as well (subject to the system parameters and depth of the object/fluid position).

(37) Upon manipulation of the valves **114**, pressure waves (e.g., a negative pressure or a positive pressure) may travel inside and along the conduits **166** and **168** through the gas **175**, the liquid **183**, or a gas-liquid interface **186**. The pressure waves may reflect off of the various fluids or interfaces within the underground storage facility **160** and may correspond to fluid levels therein. The reflected pressure waves may be received by the pressure transducers **122** and analyzed by the system controller **124** and/or the device **126**, as noted previously.

(38) The system controller **124** may be in communication and/or operable to operate various components such as the valves **114**, the pressure transducers **122**, the fluid source **182**, and/or the structure **184** via wired and/or wireless communication paths. In certain examples, the system controller **124** may operate the components based on the fluid levels within the underground

storage facility **160**.

(39) FIG. 2 illustrates a flow chart for locating and tracking of moving or stationary targets within wellbores, in accordance with examples of the present disclosure. At step **200**, at least two pressure waves may be induced within a conduit (e.g., the wellbore **102** shown on FIGS. 1A and 1B; or the underground storage facility **160** shown on FIG. 1C). As previously described, the pressure waves may be induced within the conduit by controlling flow into or out of the conduit. The pressure waves may have similar or different properties such as frequencies and/or amplitudes, in some examples. The pressure waves may travel along a bore of the conduit to the target **100** (e.g., shown on FIGS. 1A and 1B) and reflect back to an end of the wellbore **102** where the wave was induced or the source of the pressure wave inducement. The reflected pressure waves may be considered pressure responses in some examples.

(40) At step **202**, the pressure responses may be measured by a pressure transducer (e.g., the pressure transducer **122** shown on FIGS. 1A-1C) at an ultra-high sampling rate and recorded by a high frequency data recorder (e.g., the system controller **124** shown on FIGS. 1A and 1B). In some examples, the pressure waves may be induced before either pressure response is measured. In other examples, a first response may be measured before a second response is measured. The pressure responses may be analyzed on site or transmitted offsite to determine a distance of the target **100** relative to the wave inducement location.

(41) At step **204**, the preceding steps may be repeated at an interval to provide continuous real-time location updates using Equations 1-3, for example. A current location may be compared to a previous location to determine movement of the target **100**. An automated warning system (e.g., the system controller **124** shown on FIGS. 1A-1C) may send a message (e.g., text, email) to a user when a fluid level fails to satisfy a threshold or is deemed to be too high or too low.

(42) In some examples, the interval may be variable or set. A time period for each interval may include any interval longer than the time required for a wave to travel the entire length of the wellbore and return to the data recorder (e.g., the pressure transducer **122**). This may be calculated on a case-by-case basis and kept as short as possible. Location updates or location information such as fluid levels may be transmitted to a web portal to provide live tracking.

(43) FIGS. 3 and 4 illustrate pressure responses during tracking of a target, in accordance with examples of the present disclosure. As shown on FIG. 3, a pressure wave **300** may be induced to contact the target and elicit a pressure response **302** (e.g., a reflection of the pressure wave **300**) and corresponding residual data **304** such as residual pressure waves, for example. The residual data **304** may not be relied upon to track the target. As previously noted, the speed of sound in the fluid and the time from the pressure wave inducement to the first pressure response (e.g., the pressure response **302**) may be calculated. Also, a distance from a pressure transducer to the location of the target in the conduit may be calculated via Equations 1 to 3. For example, the distance to the target from the pressure transducer, may be determined as half the distance the pressure wave **300** travels from the time of the pressure wave inducement to the time the pressure response **302** is measured or received by a pressure transducer. The distance between the pressure transducer and the target may be utilized to calculate a distance to the target relative to a pressure inducement location such as a valve or pump, for example. In some examples, as illustrated on FIG. 4, measured pressures may include an induced wave **400**, noise **402**, and a pressure response **404**.

(44) In certain examples, systems and methods of the present disclosure may be applicable to either a temporary installation or a permanent installation. For the permanent installation, the system or method may be triggered by an operator manually initiating the techniques described herein with a button or software interface, for example.

(45) Accordingly, the systems and methods of the present disclosure may allow for a determination of a fluid level or a location of an object in a conduit such as a wellbore or an underground storage facility. The systems and methods may include any of the various features disclosed herein, including one or more of the following statements.

- (46) Statement 1. A system for non-intrusively monitoring a fluid level or an object in a conduit, the system comprising a component that is positioned to control flow into or out of the conduit to induce pressure waves in the conduit, wherein the conduit extends into a subterranean formation; a pressure transducer in fluid communication with the conduit, the pressure transducer positioned to measure pressure responses in the conduit due to contact of the pressure waves with the fluid level or the object; and a system controller operable to: receive pressure data from the pressure transducer, the pressure data comprising the pressure responses; and determine a distance of the fluid level or the object in the conduit, relative to the component or the pressure transducer, based on the pressure responses.
- (47) Statement 2. The system of the statement 1, wherein the pressure transducer is located at a wellhead that is in fluid communication with the conduit.
- (48) Statement 3. The system of the statement 1 or the statement 2, wherein the wellhead comprises the component.
- (49) Statement 4. The system of any one of the preceding statements, wherein a sampling rate of the pressure transducer is 1 kilohertz or higher.
- (50) Statement 5. The system of any one of the preceding statements, wherein the conduit comprises a wellbore or well annulus (pipe-in-pipe).
- (51) Statement 6. The system of any one of the preceding statements, wherein the conduit extends into an underground storage facility.
- (52) Statement 7. The system of any one of the preceding statements, wherein the underground storage facility comprises an underground cavern.
- (53) Statement 8. The system of any one of the preceding statements, wherein the underground storage facility comprises a gas.
- (54) Statement 9. A method for non-intrusively monitoring a target in a wellbore, the method comprising: controlling a component to induce at least two pressure waves in the wellbore; measuring, with a pressure transducer, pressure responses in the wellbore due to contact of the pressure waves with the target; and determining a distance of the target in the wellbore, relative to the component or the pressure transducer, based on the pressure responses.
- (55) Statement 10. The method of the statement 9, further comprising inducing at least two positive pressure waves in the wellbore via an addition of fluid into the wellbore.
- (56) Statement 11. The method of the statement 9 or the statement 10, further comprising measuring pressure responses induced by a reciprocating pump that removes fluid from the wellbore to induce at least two negative pressure waves in the wellbore.
- (57) Statement 12. The method of any one of the statements 9-11, further comprising controlling the component to allow an ingress of fluid into the wellbore or well annulus (pipe-in-pipe).
- (58) Statement 13. The method of any one of the statements 9-12, further comprising controlling the component to allow an egress of fluid from the wellbore or well annulus (pipe-in-pipe).
- (59) Statement 14. The method of any one of the statements 9-13, further comprising receiving the pressure data at intervals.
- (60) Statement 15. The method of any one of the statements 9-14, further comprising recording pressure data at a frequency ranging from 1 kilohertz (kHz) to 4 kHz, or at a frequency greater than 4 kHz or less than 1 kHz.
- (61) Statement 16. A method for non-intrusively monitoring a target in a conduit that extends into a subterranean formation, the method comprising: controlling a component to induce pressure waves in the conduit that extends into the subterranean formation; measuring pressure responses in the conduit due to interactions of the pressure waves with the target; and determining a distance of the target in the conduit, relative to the component, based on the pressure responses.
- (62) Statement 17. The method of the statement 16, further comprising controlling the component to allow fluid into the conduit to induce the pressure waves.
- (63) Statement 18. The method of the statement 16 or 17, further comprising sampling pressure

data at a rate that is at least 1 kHz.

(64) Statement 19. The method of any one of the statements 16-18, further comprising removing fluid from the conduit by controlling the component.

(65) Statement 20. The method of any one of the statements 16-19, further comprising locating hydrates based on the pressure responses.

(66) Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations may be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. The preceding description provides various examples of the systems and methods of use disclosed herein which may contain different method steps and alternative combinations of components. It should be understood that although individual examples may be discussed herein, the present disclosure covers all combinations of the disclosed examples, including, without limitation, the different component combinations, method step combinations, and properties of the system. It should be understood that the compositions and methods are described in terms of “comprising,” “containing,” or “including” various components or steps, the compositions and methods can also “consist essentially of” or “consist of” the various components and steps. Moreover, the indefinite articles “a” or “an,” as used in the claims, are defined herein to mean one or more than one of the elements that it introduces.

(67) For the sake of brevity, only certain ranges are explicitly disclosed herein. However, ranges from any lower limit may be combined with any upper limit to recite a range not explicitly recited, as well as, ranges from any lower limit may be combined with any other lower limit to recite a range not explicitly recited, in the same way, ranges from any upper limit may be combined with any other upper limit to recite a range not explicitly recited. Additionally, whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range are specifically disclosed. In particular, every range of values (of the form, “from about a to about b,” or, equivalently, “from approximately a to b,” or, equivalently, “from approximately a-b”) disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values even if not explicitly recited. Thus, every point or individual value may serve as its own lower or upper limit combined with any other point or individual value or any other lower or upper limit, to recite a range not explicitly recited.

(68) Therefore, the present examples are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The particular examples disclosed above are illustrative only and may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Although individual examples are discussed, the disclosure covers all combinations of all of the examples. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. It is therefore evident that the particular illustrative examples disclosed above may be altered or modified and all such variations are considered within the scope and spirit of those examples. If there is any conflict in the usages of a word or term in this specification and one or more patent(s) or other documents that may be incorporated herein by reference, the definitions that are consistent with this specification should be adopted.

Claims

1. A system for non-intrusively monitoring an object in a conduit, the system comprising: a component that is positioned to control flow into or out of the conduit to induce pressure waves in the conduit, wherein the conduit extends into a subterranean formation; a pressure transducer in fluid communication with the conduit containing a fluid, the pressure transducer positioned to

measure pressure responses in the conduit due to contact of the pressure waves with the object; and a system controller operable to: receive pressure data from the pressure transducer, the pressure data comprising the pressure responses from the object wherein is a cement plug moving within the conduit during a cementing operation; determine a time-log of pressure change in the conduit from the pressure data; determine a distance-log of pressure change from the time-log of pressure change and speed of sound the fluid; determine a distance of or the cement plug in the conduit, relative to the component or the pressure transducer, based on the distance-log of pressure change; track a position of the cement plug during the cementing operation by repeatedly determining said distance of the cement plug at multiple points in time as it moves through the conduit; and determine a time duration for at least a portion of the cementing operation based on the tracked position of the cement plug over said multiple points in time.

2. The system of claim 1, wherein the pressure transducer is located at a wellhead that is in fluid communication with the conduit.

3. The system of claim 2, wherein the wellhead comprises the component.

4. The system of claim 1, wherein a sampling rate of the pressure transducer is 1 kilohertz or higher.

5. The system of claim 1, wherein the conduit comprises a wellbore.

6. The system of claim 1, wherein the conduit extends into an underground storage facility.

7. The system of claim 6, wherein the underground storage facility comprises an underground cavern.

8. The system of claim 6, wherein the underground storage facility comprises a gas.

9. A method for non-intrusively monitoring a target in a wellbore, the method comprising: controlling a component to induce at least two pressure waves in the wellbore containing a fluid; measuring, with a pressure transducer, pressure responses in the wellbore due to contact of the pressure waves with the target to generate pressure data from the target, wherein the target is a cement plug moving within the wellbore during a cementing operation; determining a time-log of pressure change in the wellbore from the pressure data; determine a distance-log of pressure change from the time-log of pressure change and speed of sound the fluid; determining a distance of the cement plug in the wellbore, relative to the component or the pressure transducer, based on the distance-log of pressure change; track a position of the cement plug during the cementing operation by repeatedly determining said distance of the cement plug at multiple points in time as it moves through the wellbore; and determine a time duration for at least a portion of the cementing operation based on the tracked position of the cement plug over said multiple points in time.

10. The method of claim 9, further comprising inducing at least two positive pressure waves in the wellbore via an addition of fluid into the wellbore.

11. The method of claim 9, further comprising measuring pressure responses induced by a reciprocating pump that removes fluid from the wellbore to induce at least two negative pressure waves in the wellbore.

12. The method of claim 9, further comprising controlling the component to allow an ingress of fluid into the wellbore.

13. The method of claim 9, further comprising controlling the component to allow an egress of fluid from the wellbore.

14. The method of claim 9, further comprising receiving the pressure data at intervals.

15. The method of claim 9, further comprising recording pressure data at a frequency ranging from 1 kilohertz (kHz) to 4 kHz.

16. A method for non-intrusively monitoring a target in a conduit that extends into a subterranean formation, the method comprising: controlling a component to induce pressure waves in a fluid in the conduit that extends into the subterranean formation; measuring pressure responses in the conduit due to contact of the pressure waves with the target to generate pressure data from the target wherein the target is a cement plug moving within the conduit during a cementing operation;

determining a time-log of pressure change in the conduit from the pressure data; determine a distance-log of pressure change from the time-log of pressure change and speed of sound the fluid; determining a distance of the cement plug in the conduit, relative to the component, based on the distance-log of pressure change; and track a position of the cement plug during the cementing operation by repeatedly determining said distance of the cement plug at multiple points in time as it moves through the conduit; and determine a time duration for at least a portion of the cementing operation based on the tracked position of the cement plug over said multiple points in time.

17. The method of claim 16, further comprising controlling the component to allow fluid into the conduit.

18. The method of claim 16, further comprising sampling pressure data at a rate of 1 kilohertz or higher.

19. The method of claim 16, further comprising removing fluid from the conduit by controlling the component.

20. The method of claim 16, further comprising locating hydrates based on the pressure responses.
